VARIABLE → assignment operator

age = 27

↓ left side → variable name

→ right side → value

3 rules

→ we cannot start with numbers.

→ no space allowed.

→ no special characters.

Case Sensitive

- age ✓
- Age ✓
- AGE ✓

snake_cases — (white space)

cwlones_age

throwaway_amount

Numeric Data Types:-

↳ types

- Int (1,2,3)
- float (5.15, 4.23)
- complex (a+ib) ✓ (2 + 4i)

2 + 3 + 5 + 16 + 7i → 16 + 7i

i → iota

i = √-1

5 × 0 = 5 or 0

1000 × 0 = 0

Boolean values

↳ logical truth values → True ✓, False ✓

Boolean rules

0 → False

non-zero → True

empty string → False

non-empty " → True

Arithmetic operators

Addition +	2+3 = 5	2×2×2 =
Subtraction −	5−3 = 2	
Multiplication × (*)	5×3 = 15	
Division /	10/2 = 5	10³/3 → 3.331
Modulus %	10%3 = 1 → remainder	
Exponent **	2**3 = 8	
Floor division //	10//3 = 3	

Comparison operators =

== → equals

!= → not equal

5 > 3 ↳ Greater than

5 < 3 ↳ less than

5 <= 3 → False

5 <= 5 → True

Text Data

String → sequence of characters enclosed with quotes

↓ letters, numbers, symbols, spaces.

This class is amazing.

↳ review = "This class is amazing"

String length

len() → text = "Mahipal"

print (len(text))

⇒ 7

name = "Artificial Intelligence"

print (len(name))

⇒ 23

String Indexing

text = "Python"

0	1	2	3	4	5
P	y	t	h	o	n
-6	-5	-4	-3	-2	-1

(−0)

0 → zero indexing

← negative indexing

Slicing

"Mahipal" → "pal"

string → substring ✓

inclusive → [start : end] ← exclusive

Step Slicing

[start : end : step]

```
word = "Machinelearning"
print(word[0:15:2])
```

MCIEERIG

String Immutability

```
text = "pyton"
text[0] = "J"
```

error. ✓

```
text = "pyton"
new_text = "J" + text[1:]
print(new_text)
```

Jython.

M-1: Concatenation

```
name = "Tommy"
salary = 50000
print("Hello" + name + "your salary is" + str(salary))
```

↳

M-2: f-string

```
name = "Tommy"
salary = 50000
print(f"Hello {name}, your salary is {salary}")
```